

数字信号处理

Digital Signal Processing

第一讲：绪论

电子工程系 刘一民

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罗姆楼11-102

本讲提纲

1. 基本信息

1. 教师信息
2. 课程信息与教学计划
3. 考核方式

2. 引言

1. 历史与现在
2. 知识体系与基本概念

3. 采样——从模拟到数字

1. 采用的过程
2. 采样定理
3. 近期的研究: 信息采样

1. 基本信息

1. 教师信息

1. 刘一民：长聘教授，信息系统研究所（信检）

1. 研究方向：Signal Processing, Compressive Sensing and Sparse Recovery, Radar and other active sensing technologies.

2. 答疑安排：

1. 周一下午2:00~3:30,

2. 罗姆楼11-102（建议提前邮件确认下是否出差...）

2. 助教：

郑子恒, zhengzh22@mails.tsinghua.edu.cn,
罗姆楼11-109, 手机号18611663766

1. 基本信息

1.1 课程信息

- 课程目标

- ✓学习、理解和掌握数字信号处理的基本理论和方法

- 课程教材

- ✓张旭东 等, 《数字信号分析和处理》(第二版), 清华大学出版社

- 先修课程

- ✓微积分、信号与系统、线性代数、复变函数、(离散数学)

- 参考书

- ✓A. V. Oppenheim, R. W. Schaffer, Discrete-Time Signal Processing (3rd edition), 2011, Pearson Education.

- ✓J. Proakis, Digital Signal Processing

- ✓Matlab® Help

1. 基本信息

1.2 教学计划

教学内容	学时数
课程介绍和概念、知识回顾	3
离散信号和系统的表示	3
DFT: 概念性质、快速计算和应用	9
数字频谱估计	3
DCT、离散正交变换及其他	3
离散LTI系统的分析和设计	12
有限字长效应	3
采样率变换、压缩感知及其他	3
总结与回顾	3

1. 基本信息

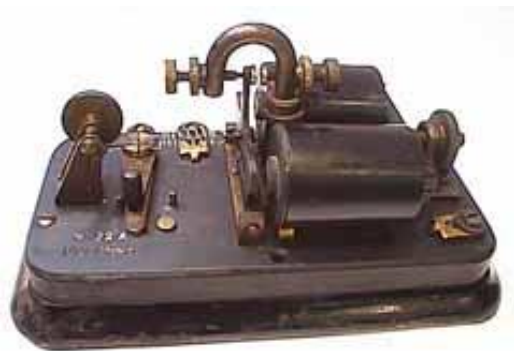
1.3 考核方式

- 作业 (35%)
 - 平时作业 (20%) (需线下提交)
 - 大作业 (一次) (15%)
- 期末考试 (65%)
 - 开卷
 - 可携带
 - 考试工具
 - 教材
 - 讲义, Slices
 - 笔记
 - 不能携带任何可上网的工具

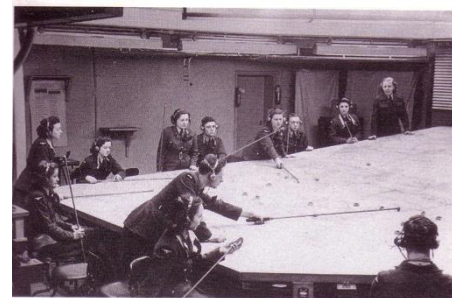
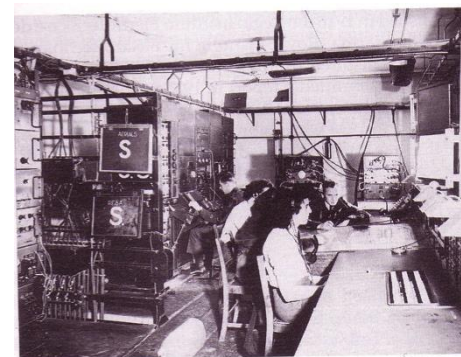
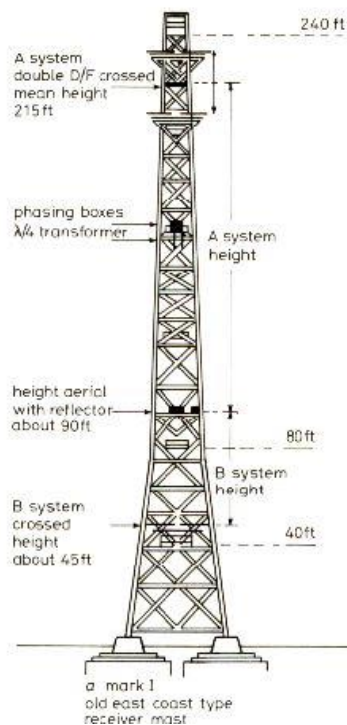
2. 引言

20世纪早期

通信

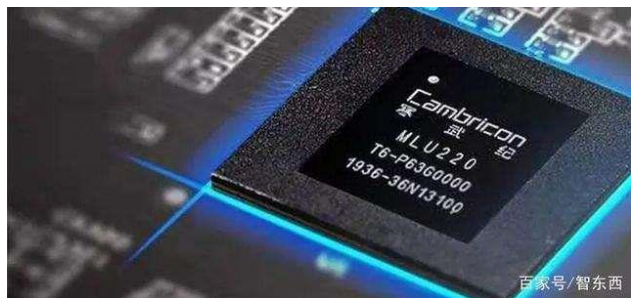
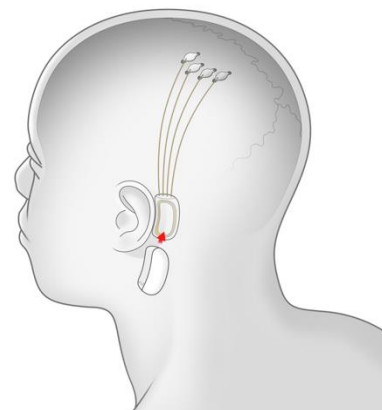


雷达

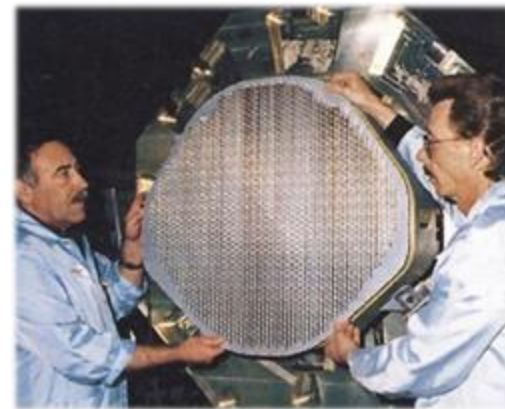


2. 引言

现在



百家号/智东西



2. 引言

2.2 信号处理的概念和意义

- 什么是信号

- A signal is a physical quantity which varies with respect to time, space and contains information from source to destination.

确定性“消息”
不含信息

- Signals are represented mathematically as functions of one or more independent variables.

Examples?

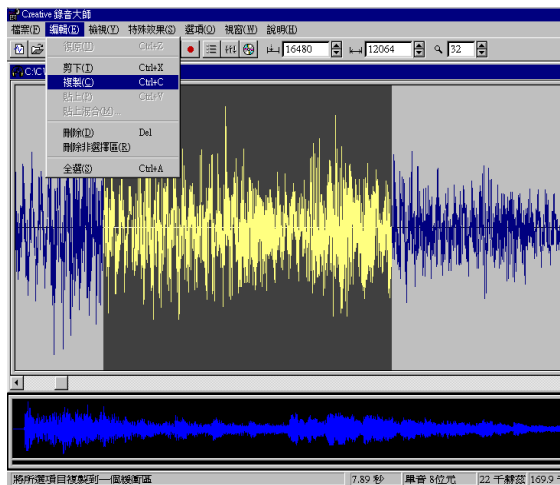
- From an engineering point of view, electronic signals are more preferable.

Why? or?

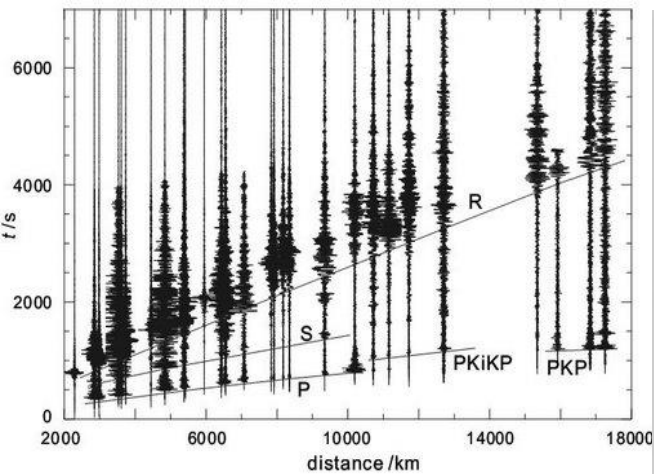
2. 引言

信号的例子

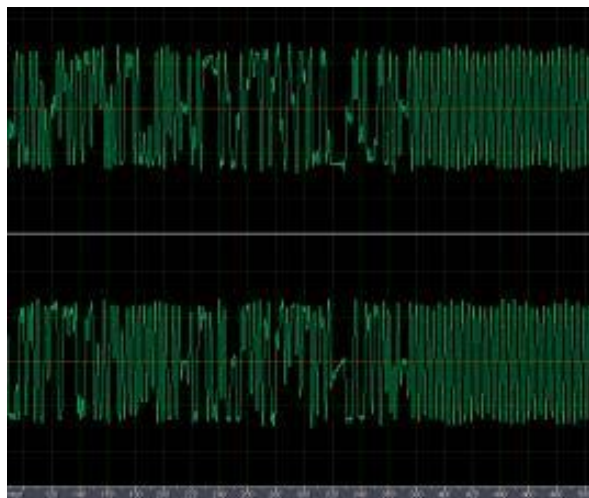
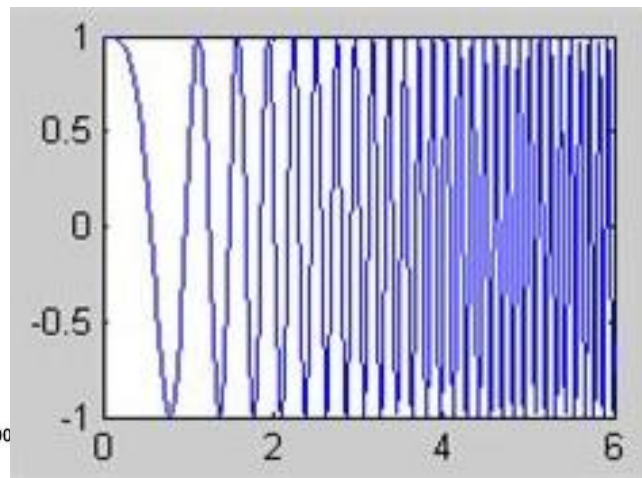
Voice



Earthquake



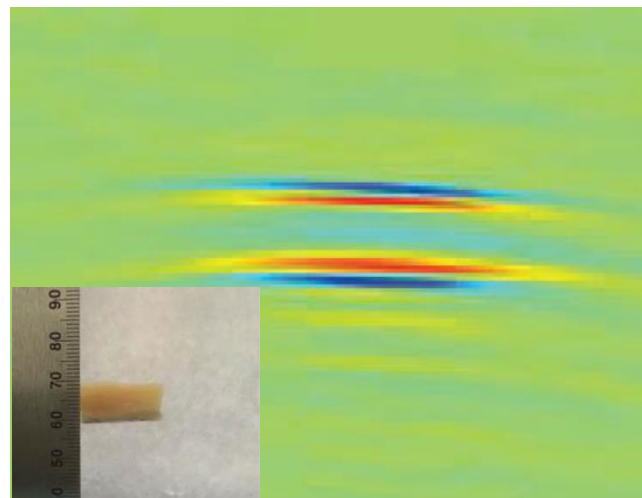
Radar



Communication



image

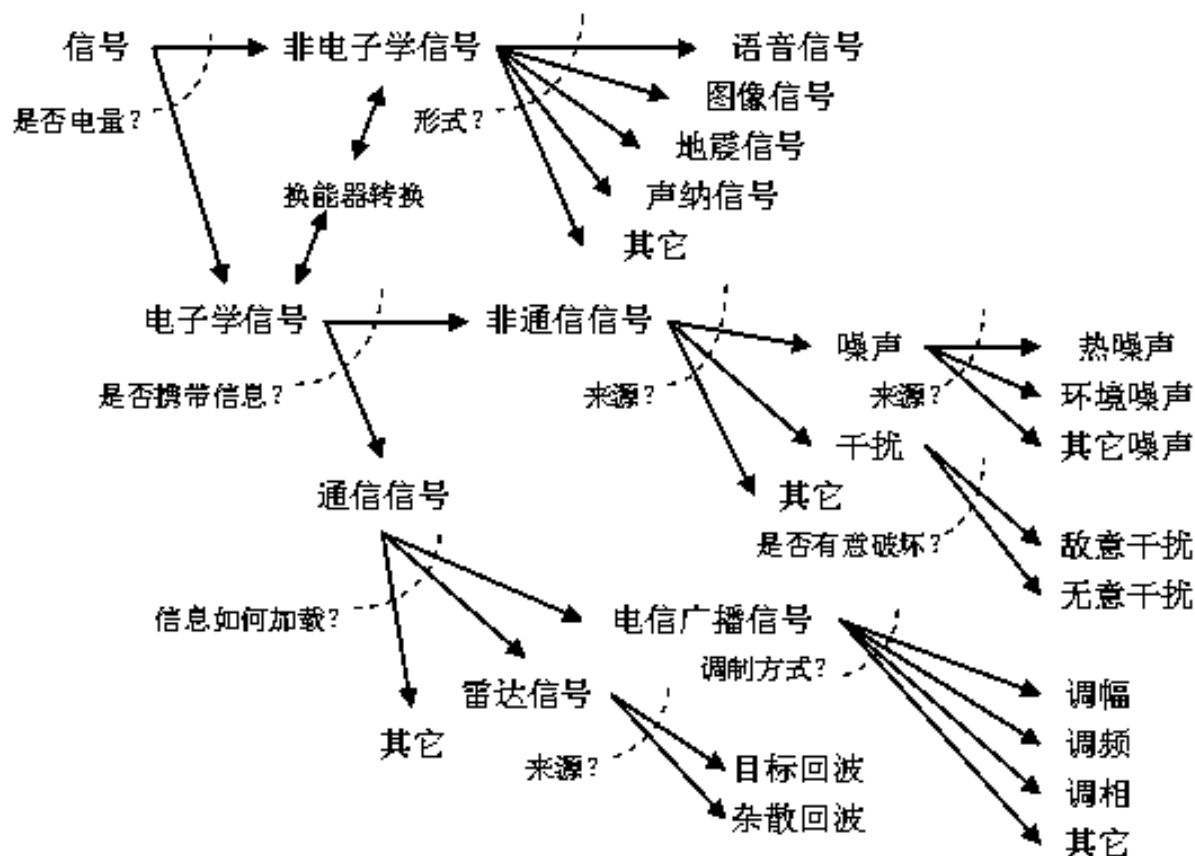


Biomedical

2. 引言

2.2 信号处理的概念和意义

- 信号的分类



问题:

1. 信号一定是单变量函数?
2. 除时间以外, 信号还可以是那些变量的函数?

2. 引言

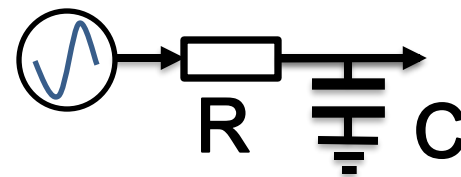
2.2 信号处理的概念和意义

- “信号处理” 的概念
 - 研究信号及其所蕴涵的**信息的表示**(Representation)、**变换**(Transform)和**操作**(Manipulation)。

- 表示--例：Image/Video Coding

- 变换--例：Fourier Transform $\int_{-\infty}^{+\infty} f(t)e^{-j\omega t} dt$

- 操作--例：滤波



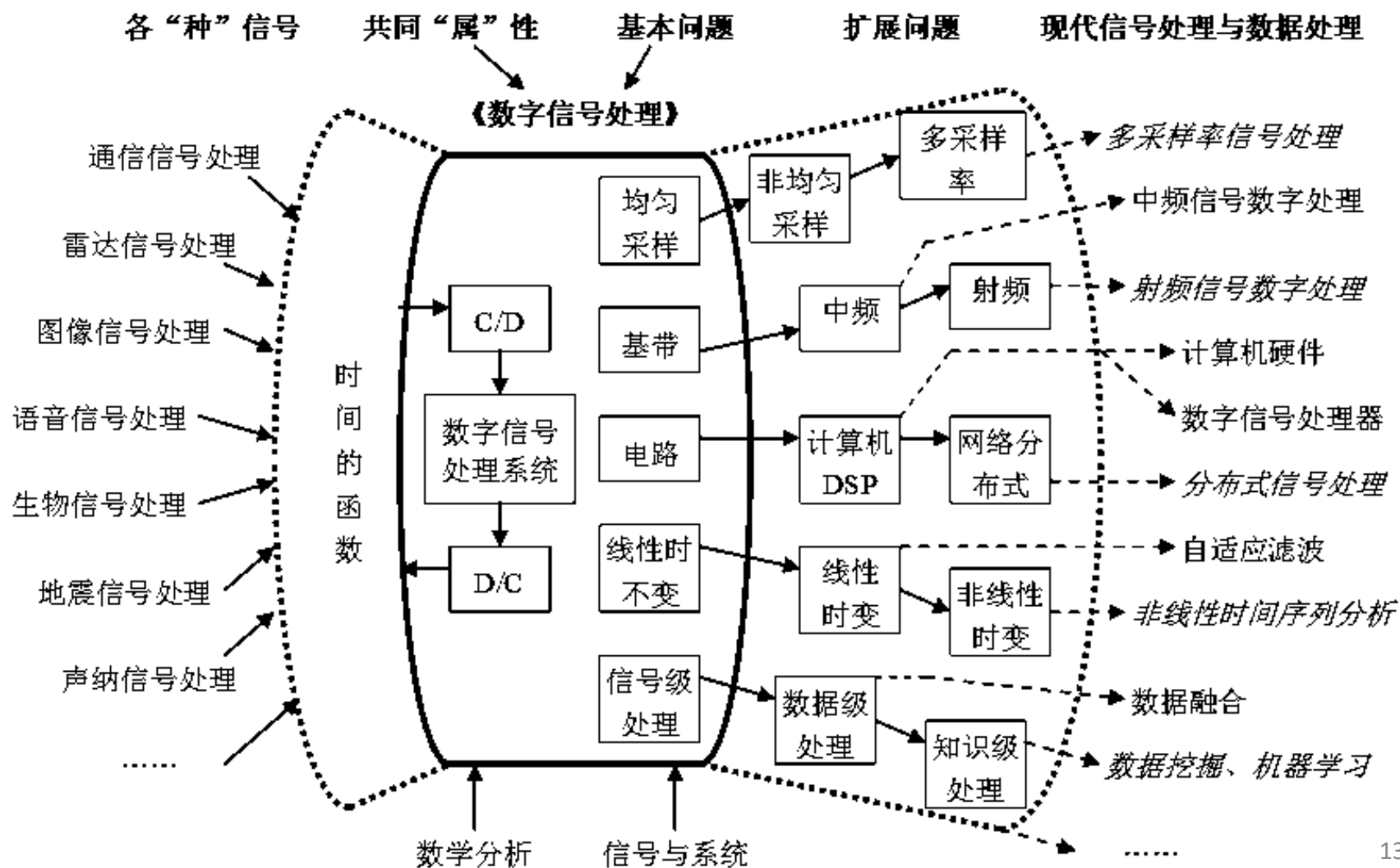
- 信号处理是涉及**数学**、**物理**和**工程**的综合问题

- 例：傅立叶变换

- 数学：正交变换、本征分析
- 物理：谐波分析
- 工程：滤波器组

2. 引言

- 在知识体系中的位置



2. 引言

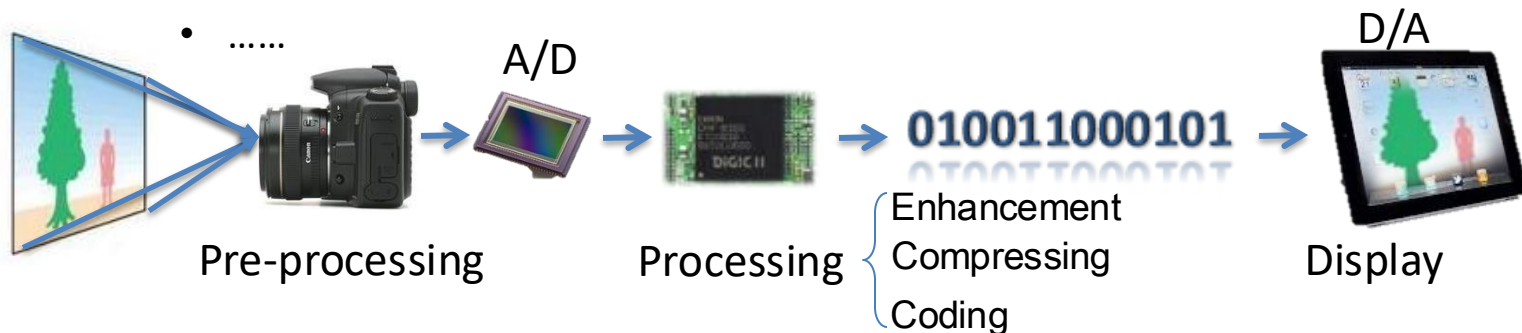
- 数字（离散时间）信号处理的特点

- Pros

- **Accuracy** can be controlled
 - **Flexibility** with software implementations
 - **Complex** systems and algorithms can be implemented
 - Sensitivity to electrical noise is minimal
 - Digital storage is **cheaper**
 - Digital information can be **more easily** encrypted and compressed
 -

- Cons

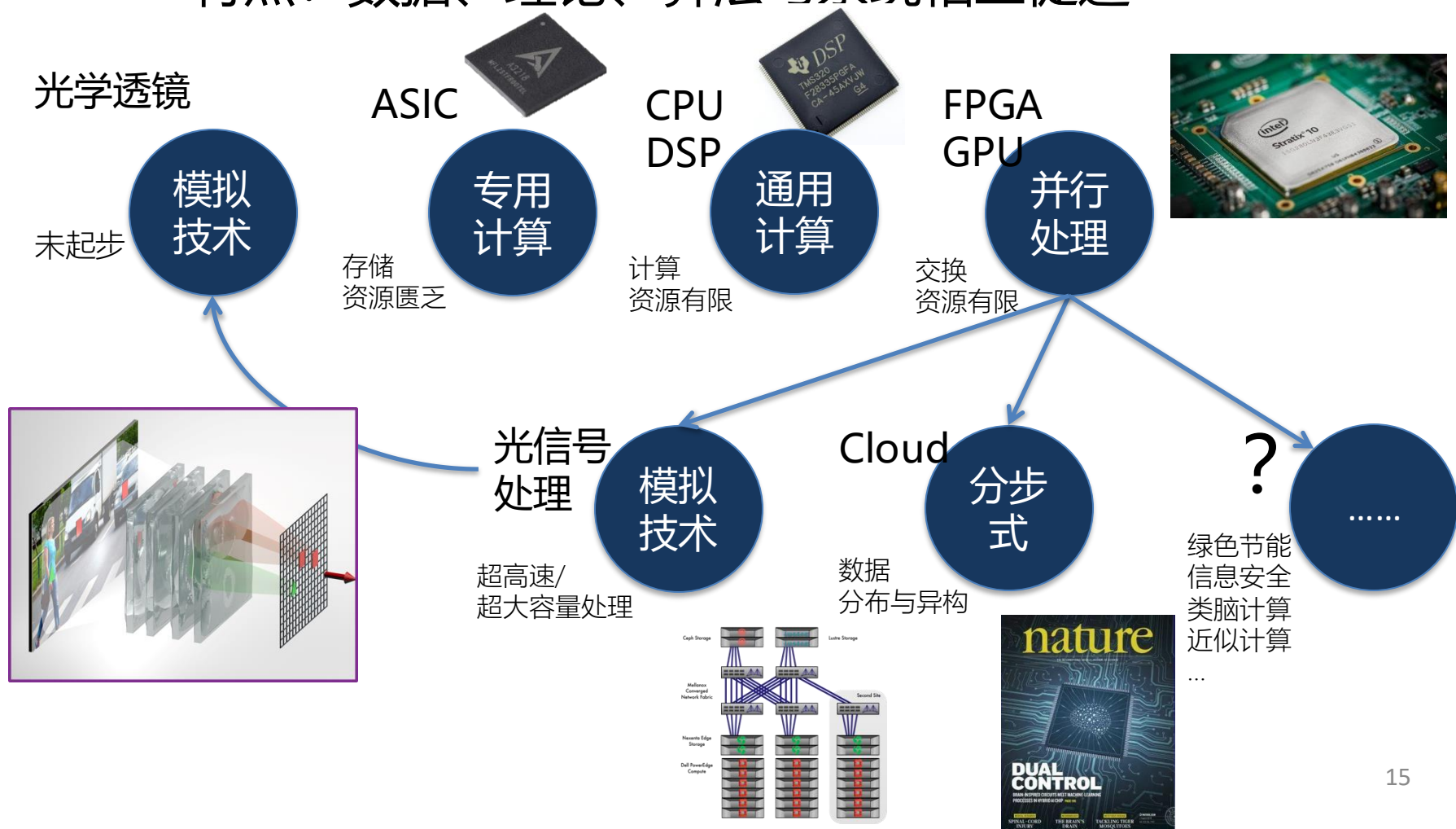
- Limitations of A/D and D/A
 - Quantization and round-off errors
 - Limited to signals with **relatively** low bandwidths
 -



2. 引言

- 学科发展历史

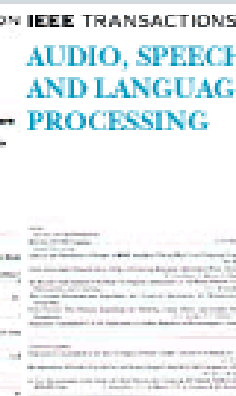
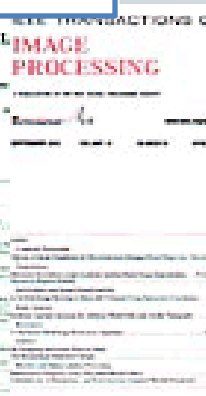
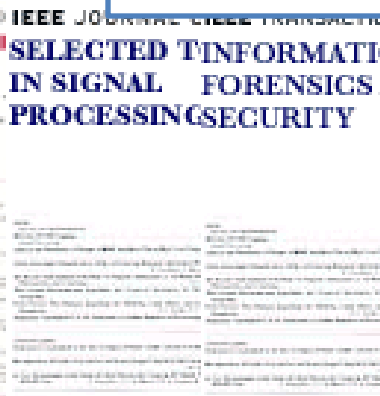
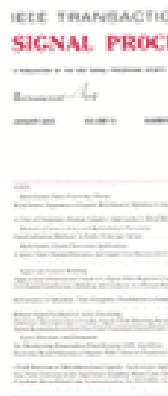
- 特点：数据、理论、算法与系统相互促进



2. 引言

“信号处理”的研究领域

Signal Processing Theory;
Algorithms;
Communication;
Radar/Sonar/Navigation;
Image/Video;
Audio/Speech/Language;
Smart grid;
Biomedical;
Computational Imaging;
Autonomous Driving;
A.I.;
FinTec...



Sponsored/Co-Sponsored Conferences

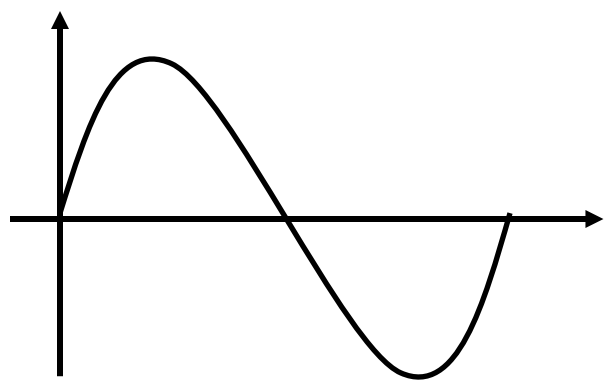
Upcoming Conferences	Call-for-Paper Submission Deadline	Location	Conference Dates
10th IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS 2013)	March 25, 2013	Krakow, Poland	August 27-30, 2013
2013 IEEE International Conference on Image Processing (ICIP 2013)	February 5, 2013	Melbourne, Australia	September 15-18, 2013
2013 IEEE International Workshop on Machine Learning for Signal Processing (MLSP)	June 5, 2013	Southampton, United Kingdom	September 22-25, 2013
2013 IEEE 15th International Workshop on Multimedia Signal Processing (MMSP 2013)	April 12, 2013	Pula, Italy	Sept. 30-Oct. 2, 2013
2013 IEEE Workshop on Signal Processing Systems (SIPS 2013)	April 28, 2013	Taipei, Taiwan	October 15-18, 2013
2013 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA 2013)	May 17, 2013	New Paltz, NY	October 20-23, 2013
2013 IEEE International Workshop on Genomic Signal Processing and Statistics (GENSIPS 2013)	August 2, 2013	Houston, TX	November 17-19, 2013
2013 IEEE International Workshop on Information Forensics and Security (WIFS 2013)	June 30, 2013	Guangzhou, China	November 18-21, 2013
IEEE Global Conference on Signal and Information Processing (GlobalSIP 2013)	June 15, 2013	Austin, TX	December 3-5, 2013
2013 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU 2013)	July 8, 2013	Olomouc, Czech Republic	December 8-12, 2013
2013 IEEE 5th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP 2013)	July 26, 2013	Saint Martin	December 15-18, 2013
2014 IEEE World Forum on Internet of Things (WF-IoT)	September 30, 2013	Seoul, South Korea	March 6-8, 2014
2014 IEEE International Symposium on Biomedical Imaging (ISBI 2014)	October 1, 2013	Beijing, China	April 28-May 2, 2014
2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2014)	October 27, 2013	Florence, Italy	May 4-9, 2014
The Eighth IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM 2014)	January 10, 2014	A Coruña, Spain	June 22-25, 2014
2014 IEEE Statistical Signal Processing Workshop (SSP 2014)	TBD	Gold Coast, Australia	June 29-2 July, 2014
2014 IEEE International Conference on Multimedia and Expo (ICME 2014)	December 9, 2013	Chengdu, China	July 14-18, 2014
2014 IEEE International Conference on Image Processing (ICIP 2014)	January 31, 2014	Paris, France	October 27-30, 2014
2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2015)	TBD	Brisbane, Australia	April 19-24, 2015
2015 IEEE International Conference on Image Processing (ICIP 2015)	TBD	Quebec City, Quebec	Sept. 28-Oct. 1, 2015
2016 IEEE International Conference on Image Processing (ICIP 2016)	TBD	Phoenix, Arizona	September 25-28, 2016
2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)	TBD	New Orleans, LA, USA	March 5-9, 2017

3. 采样——从模拟到数字

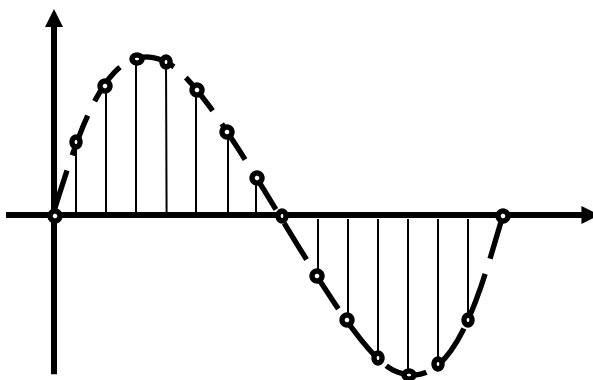
3.1 信号的分类

		定义域	
		连续	离散
值域	连续	连续信号	离散(时间)信号
	离散	?	数字信号

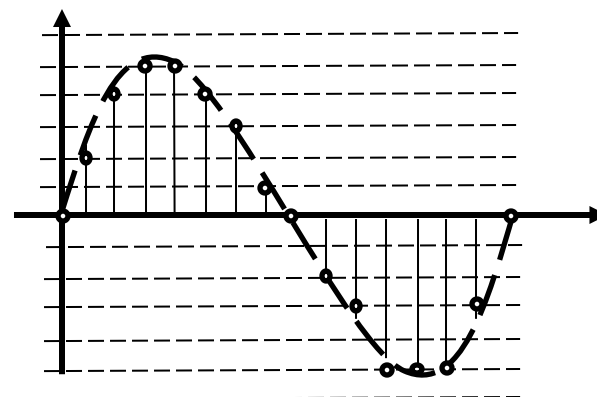
用 $x(t)$ 表示连续信号，用 $x[n]$ 表示离散时间信号



连续信号



离散(时间)信号



数字信号

3. 采样——从模拟到数字

- 离散时间信号的来源

- 定义域非连续

- 能级跃迁

- 阵列天线

- 天线阵元的位置坐标: $nd, n=0,1,\dots,N-1$

- Direction of Arrival (DOA) 估计问题

- 从连续信号中采样得到

- $x[n]=x(nT)$

- T 为采样间隔

- $[\cdot]$ 为离散信号, $(\cdot)$ 为连续信号

- 例: $x[n] = \sum_n x(nT) \cdot \delta(t - nT)$

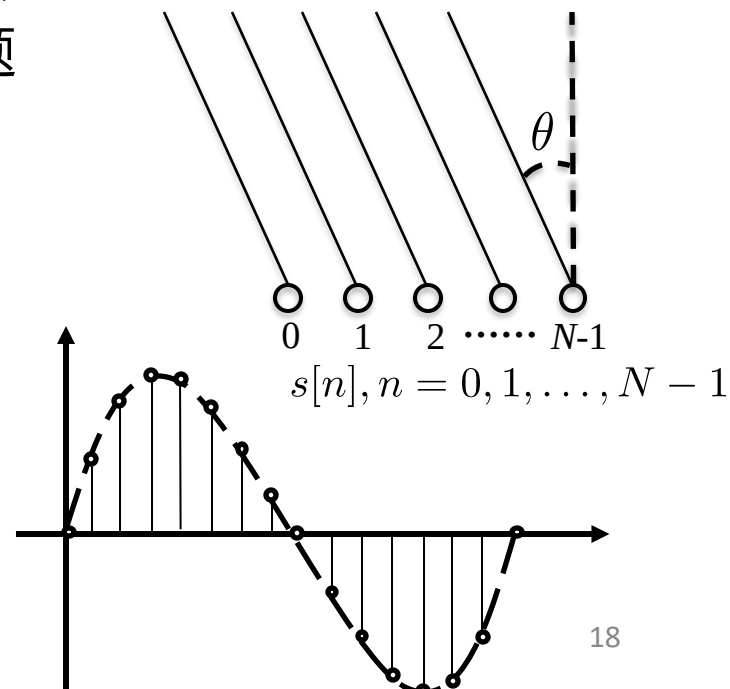
- 注意区分

- 离散时间信号 $\leftrightarrow$ 连续信号

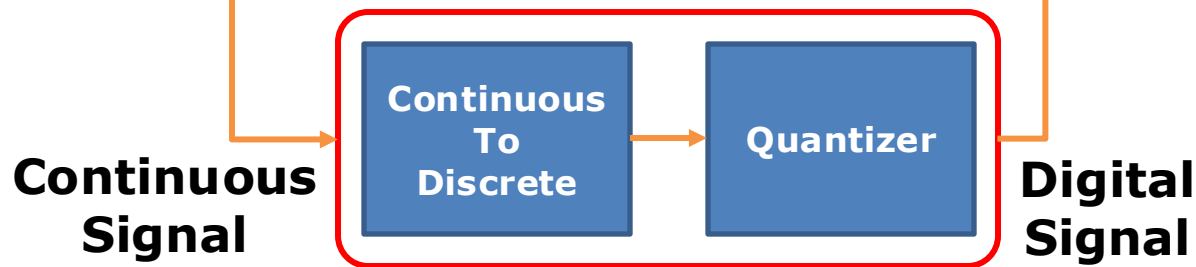
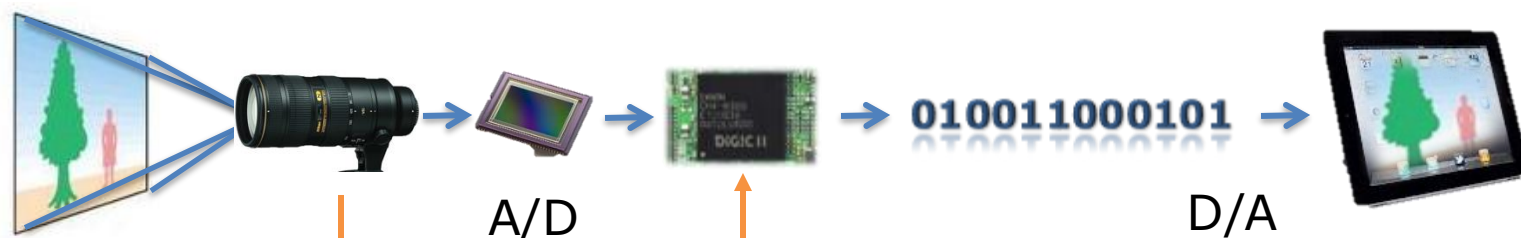
定义域: 连续
 $x(t)$



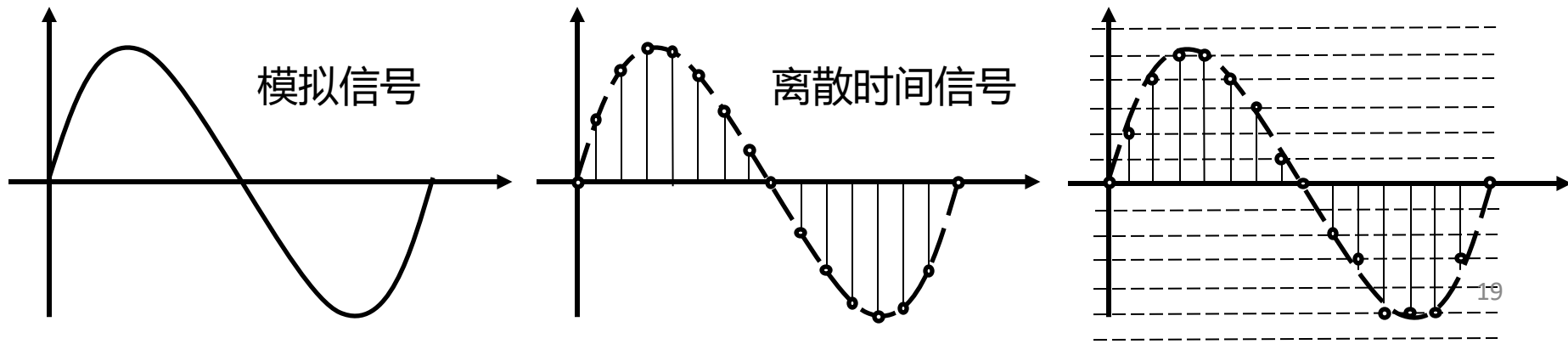
定义域: 离散
 $x[n]$



3. 采样——从模拟到数字



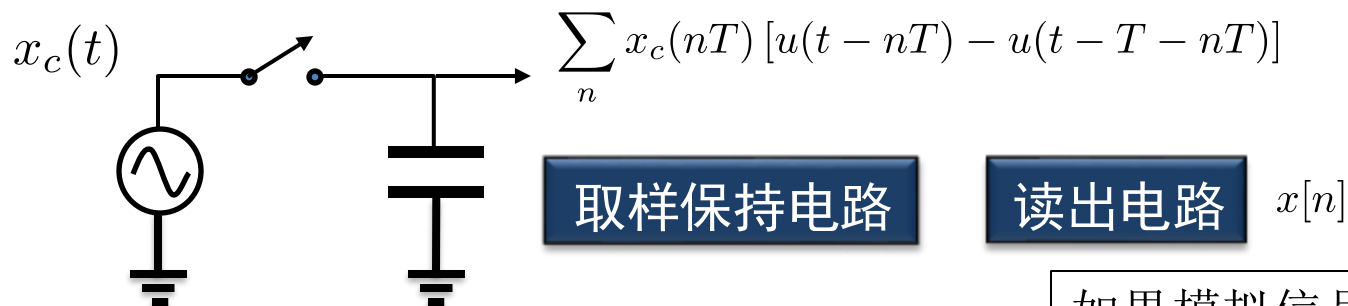
数字信号



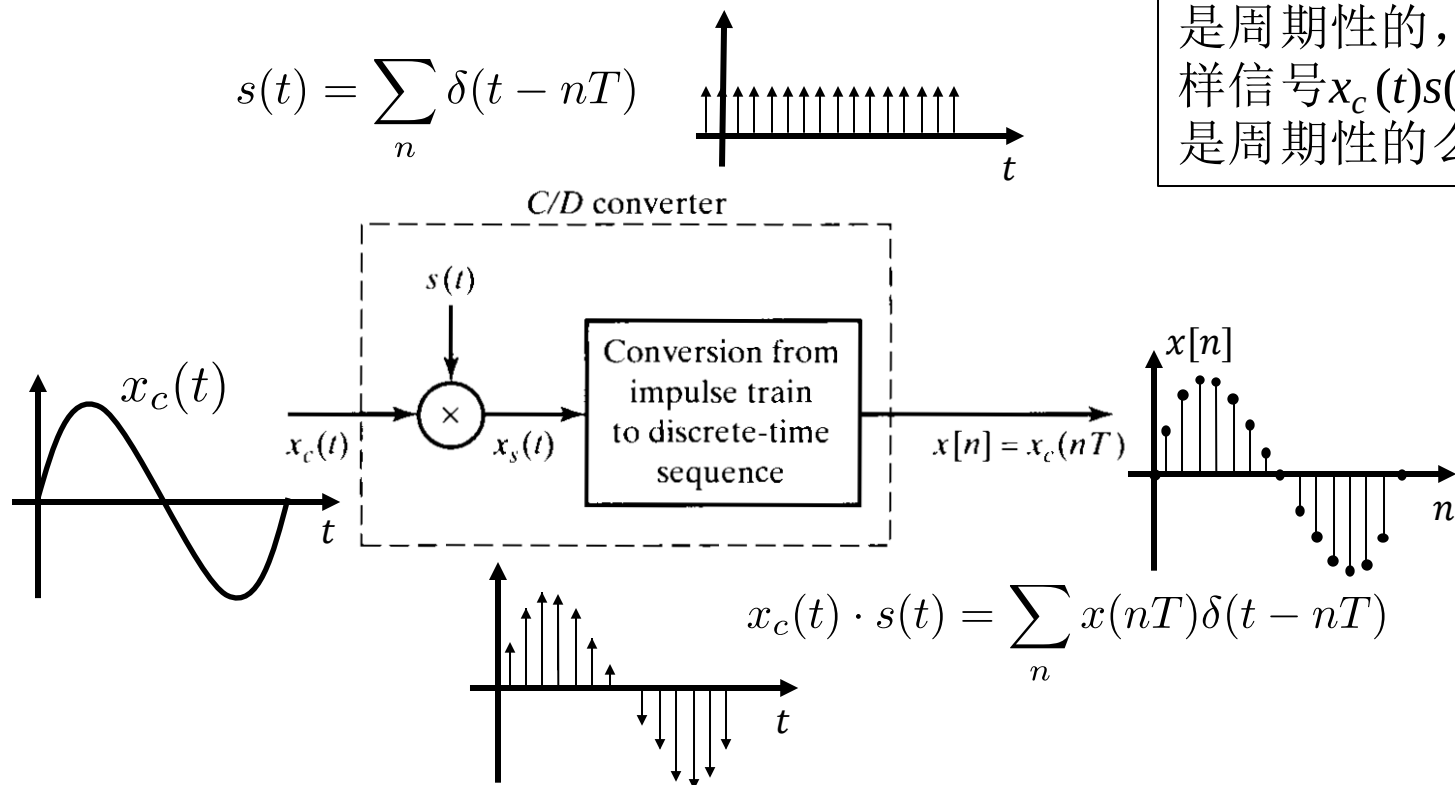
3. 采样——从模拟到数字

3.2 采样的数学表示

模拟实现



数学表达

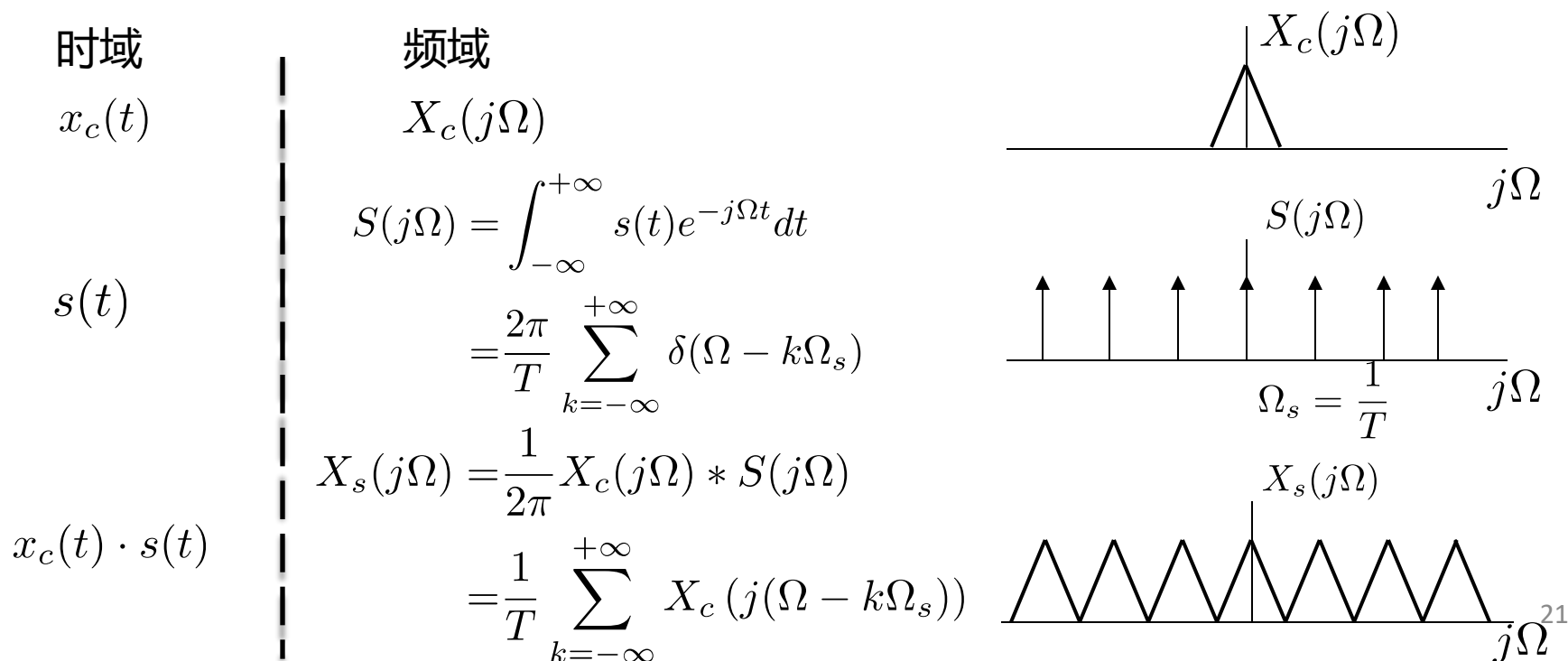
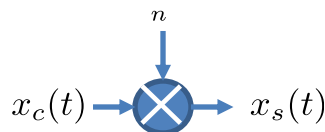


3. 采样——从模拟到数字

3.2 采样的数学表示

- 《信号与系统》：对连续信号进行采样，其频谱是原始信号的周期延拓，延拓周期为采样频率。

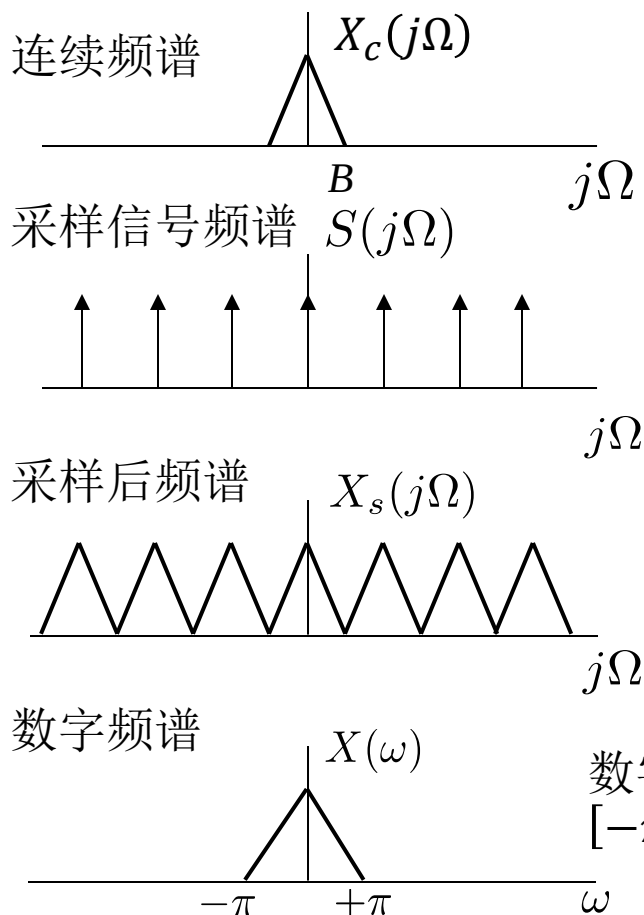
$$s(t) = \sum_n \delta(t - nT)$$



3. 采样——从模拟到数字

3.3 Nyquist采样定理

– 基本出发点：频谱不混叠，信号能恢复



假设 $x_c(t)$ 是带限信号，即

$$X_c(j\Omega) = 0, \quad \text{if } |\Omega| \geq 2\pi B$$

那么，从 $x[n]$ 中无模糊的恢复出 $x_c(t)$ 的充分条件是：采样率

$$\Omega_s = \frac{2\pi}{T} \geq 4\pi B$$

注意，上述推导中：

1. $x_c(t)$ 是基带、带限信号
2. Nyquist 是充分条件，对被采样信号无其他假设。

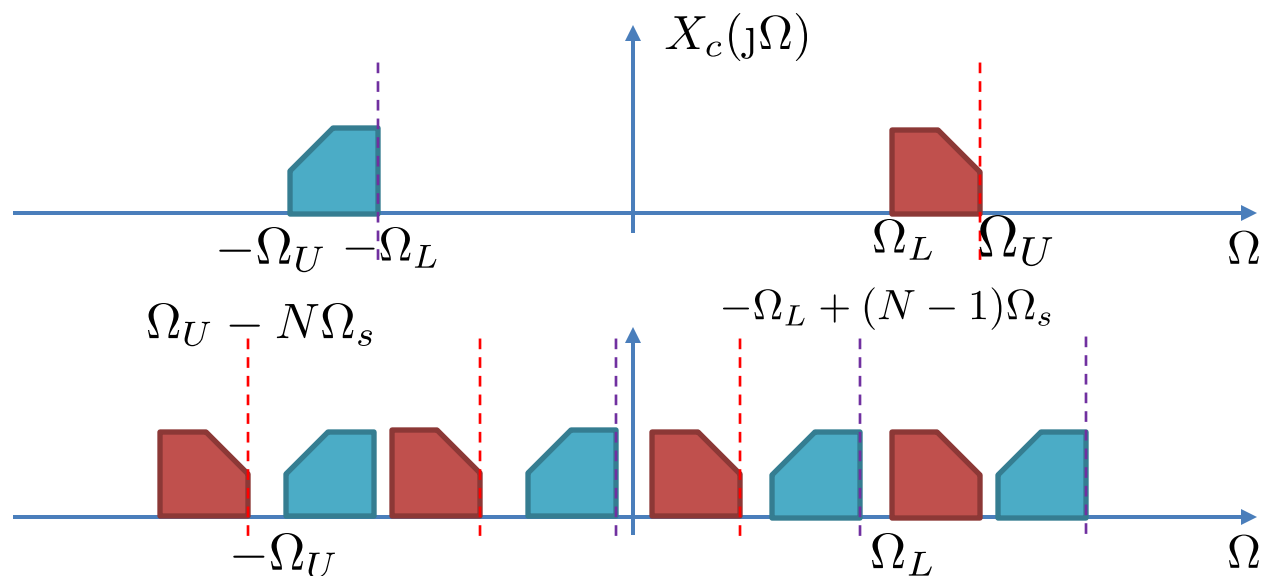
如果这两个条件不被限制？

Nyquist: 能否从数字频谱中恢复出连续频谱？

3. 采样——从模拟到数字

3.3 Nyquist采样定理

- 第一个不满足：非基带、带限信号——带通采样定理



$$\begin{cases} \Omega_U - N\Omega_s \leq -\Omega_U \\ -\Omega_L + (N-1)\Omega_s \leq \Omega_L \end{cases} \quad \longrightarrow \quad 2\Omega_U/N \leq \Omega_s \leq 2\Omega_L/(N-1)$$

带通采样定理

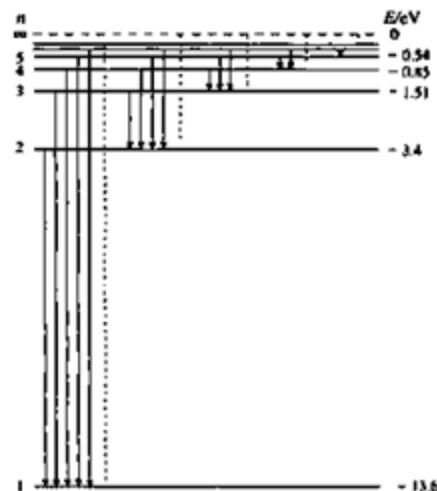
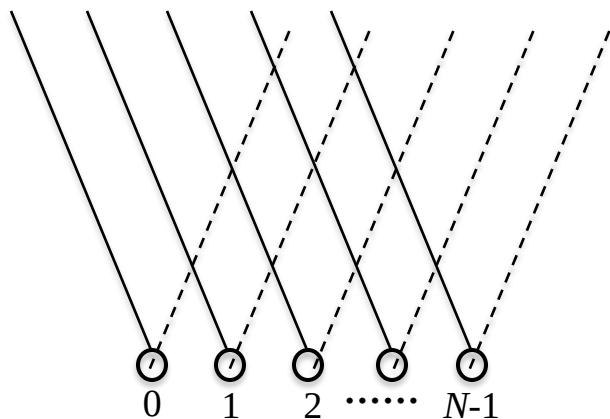
3. 采样——从模拟到数字

3.4 信息采样

– 第二个不满足：对信号有一定“先验”假设

Nyquist定理是充分条件而非必要条件，即除了带限外，信号可以是任意形态。但是实际中我们经常可以对信号有一定的先验（prior）假设。

例如，在DOA估计中，信号一般是从少数几个方向入射的（辐射源个数少）；氢原子的能谱只有少数几个能级，等等。



3. 采样——从模拟到数字

- 从另一个角度看待采样

- 不是简单的恢复信号，而是恢复“信息”本身

假设我们已知信号 $x(t)$ 是由 N 个分量

$$s_1(t), s_2(t), \dots, s_N(t)$$

中的不多于 K 个分量线性组合而成，即

$$x(t) = \sum_{i=1}^K \alpha_i s_i(t)$$

那么采样的意义在于得到 α_i ，而非直接恢复 $x(t)$ ！

思考： $s_i(t)$ 的哪些特性更有利于提高估计 α_i 的准确性？

$$\begin{bmatrix} x[1] \\ \vdots \\ x[T] \end{bmatrix} = \begin{bmatrix} s_1[1] & \cdots & s_N[1] \\ \vdots & \ddots & \vdots \\ s_1[T] & \cdots & s_N[T] \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{bmatrix}$$

矩阵S的性质

小结与作业

- 知识点:
 - 阅读《张》0.1~0.5, 《O》1, 4.1-3;
 - 信号、信号处理与数字信号处理的概念
 - 连续信号、离散时间信号、数字信号的定义
 - 连续信号与离散时间信号的关系
 - 时域
 - 频域
 - 频率与数字频率
 - 傅立叶变换与DTFT
 - 采样
 - 基带采样 (Nyquist定理)
 - 带通采样
 - 信息采样
- 作业
 - Oppenheim (3rd ed.) : 4.2, 4.3 4.4, 4.21, 4.25;
 - 《张》 : 1.4, 1.7

下集预告

- 《张》书第1.1, 1.2, 1.3, 2.1 2.2, 2.3小节